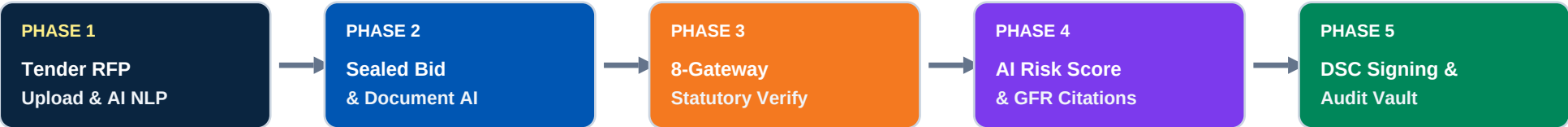


GeM-CompliAI: Master Workflow & Operational Architecture

Complete Technical Pipeline Diagram • GFR-2017 Regulatory Gateways • 3 User Workspaces



1. Phase-by-Phase Technical & Statutory Execution Pipeline

Phase & Lead Actor	Step 1 & Step 2 Operations	Automated AI & Data Transformations	Statutory Checkpoints & Output Proofs
Phase 1: Tender Formulation <i>Buyer Desk / Officer</i>	- Upload RFP / Tender Notice (PDF) - Parse Eligibility Requirements	- NLP LLM Model extracts Turnover thresholds, EMD exemptions, MSME & MII local content clauses. - Generates structured dynamic checklist.	Output: Standardized eligibility checklist rules (JSON) published to GeM Tender Registry.
Phase 2: Bid Submission <i>Tender Seller / OEM</i>	- Run Self Pre-Check Tool - Submit Sealed Bid Package	- High-accuracy Document AI (OCR) scans uploaded balance sheets, GST certificates & CA proofs. - Extracts UDIN numbers & authorized dates.	Output: Encrypted Bid Envelope (AES-256-GCM) with extracted OCR text & parsed metadata.
Phase 3: 8-Gateway Sync <i>AI Verification Engine</i>	- Synchronous API querying - Cross-Reconcile Claimed vs Verified	GSTN: Returns 3-Yr turnover & active status. MCA21: Checks CIN, Directors & balance sheets. MSME Udyam: Verifies MSE investment cap. ITD PAN: Validates corporate entity & ITRs. EPFO: Confirms active staff headcount. CVC Debarment: Checks blacklist orders.	Output: 8 Proof Dossier Cards highlighting exact matches or numerical discrepancies.
Phase 4: Risk & Reasoning <i>AI Statutory Engine</i>	- Multi-Vector Risk Calculation - Generate Legal Explanations	- Calculates weighted risk index (0–100 scale). - Assigns risk band: Low (0-20), Moderate (21-40), High (41-70), Critical (71-100). - Generates legal reasoning with GFR citations.	Output: Natural-language legal evaluation report citing GFR Rules 144, 151, and 175.
Phase 5: Decision & Vault <i>Procurement Officer</i>	- Inspect Side-by-Side Dossier - Apply Class-3 DSC Digital Seal	- Officer records binding decision (Qualify / Disqualify / Seek Clarification). - Cryptographic digest generated with officer PKI token. - Appends record to immutable SHA-256 ledger.	Output: QR-Coded Compliance Certificate + Permanent Audit Vault entry.

2. User Workspace Swimlanes & Functional Pathways

User Role & Landing Workspace	Step 1: Entry & Discovery	Step 2: Core Operational Action	Step 3: Verification / Inspection	Step 4: Final Settlement / Seal
■■ Procurement Officer Landing: AI Compliance Engine	Accesses Bidder Matrix on DOSSIER tab to view all participating tenderers.	Drills down into individual Bidder Dossier with OCR text & CA certificate UDIN proof.	Reviews 8-portal cross proofs & Explainable AI citations (AI_EXPLANATION).	Signs binding qualification order with Class-3 DSC Token ; logs to Audit Vault.
■ Buyer Desk (Indenting) Landing: Marketplace Catalog	Browses GeM e-Marketplace catalog with MSME & MII Class-1 filters.	Creates Direct Purchase PO (under ■25k) or issues RFP tender notice.	Inspects physical goods delivery at consignee premises against specs.	Digitally signs CRAC Certificate within 10 days to release GPA escrow funds.
■ Seller / Supplier Desk Landing: Seller & OEM Hub	Browses active tenders on SELLER_PORTAL & Reverse Auctions on BIDS_RA.	Runs automated Self Pre-Check tool to test profile against RFP criteria.	Submits sealed electronic bid package with tax & corporate proofs.	Participates in live Reverse Auction decrements & downloads compliance certificates.

3. Technical Architecture, Cryptography & Data Flow

Architecture Layer	Components & Technology Stack	Security & Governance Implementation
Presentation Layer (Frontend)	- React 19 Single Page App with Vite Bundler - Role-Adapted Mega Navigation Bar 14-Step Interactive Stepper Component - Real-Time Reverse Auction Decrement Arena - Vanilla CSS Government Design System Tokens	- Strict Role-Based View Interception (GFR Rule 144 Competitor Isolation) - Web Crypto API (SubtleCrypto) client-side envelope decryption - Zero plain-text persistence in browser localStorage
Application & AI Layer (Backend)	- FastAPI (Python 3.13) Asynchronous REST Framework - Multi-Vector Risk Index Scoring Engine (0–100) - Statutory Citation Generator (GFR 2017 Rules Engine) - OCR & UDIN Parser Subsystem	- Microsecond API response latency via async connection pooling - JSON Web Token (JWT) with Aadhaar OTP 2FA simulation - Strict CORS origin headers & input sanitization
Persistence & Audit Vault	- SQLite3 Relational Database Engine - SQLAlchemy ORM (Tenders, Bids, Verifications, Orders) - Append-Only Audit Trail Table (audit_logs)	- SHA-256 Cryptographic Digest verification on every row write - Automated database backup & reset administrative endpoints - Digital Certificate PKCS#7 non-repudiation sealing